Parameters Transfer, Policies May Not: Diagnosing Source Static Degeneracy in Cross-Asset Decision Gating

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Abstract

We identify Source Static Degeneracy, a silent failure mode in cross-asset decision-layer transfer where a frozen contextual policy degenerates into a fixed probability threshold. This degeneracy remains invisible to standard backtesting protocols: the collapsed policy may still outperform a naive static threshold, masking the complete loss of contextual adaptation. A quantitative team deploying such a policy would conclude that "transfer works," unaware that the policy has abandoned context-dependent decision-making entirely.

Across 228 source-target pairs spanning seven asset categories, we establish an empirical taxonomy: degenerate sources (SPY, QQQ, EEM, GLD) trigger universal action concentration on 100% of targets, whereas diverse sources (TLT, IWM) maintain behavioral variation. We document a strong empirical association between source-side action concentration and target-side degeneracy, providing sufficient conditions linking source diversity to transfer outcomes.

We introduce FCTT-DA (Frozen Contextual Threshold Transfer with Degeneracy Audit), a deployable framework featuring source diversity pre-checks, behavioral-equivalent baseline comparisons, and transaction cost sensitivity analysis. This framework is designed to detect what standard backtests cannot: policies that appear to adapt but are behaviorally static.

Our findings challenge the implicit assumption that contextual policies inherently transfer contextual behavior. The critical question is not whether a policy transfers, but whether it transfers adaptation. FCTT-DA provides the diagnostic tools to answer this question prior to deployment. In our experimental setting, source-side action concentration proves to be a highly effective screening diagnostic for transfer failure.

Keywords: *contextual bandits, policy transfer, decision gating, source static degeneracy, cross-asset learning*

1. Introduction

1.1 The Operational Appeal of Decision-Layer Reuse

In quantitative asset management, the standard workflow separates prediction from decision. A machine learning model forecasts the probability of a favorable outcome, and a decision layer converts that probability into a trading signal. While asset-specific predictors are now common, optimizing and validating a separate decision layer for every target asset is computationally costly and statistically fragile. This creates a strong operational incentive: train a decision policy on a highly liquid, well-studied source asset and deploy it, frozen, across multiple targets.

Consider a concrete example: a quantitative team develops an optimal decision policy for the SPDR S&P 500 ETF (SPY) that achieves an annualized Sharpe ratio of *0.45* over five years of walk-forward validation. The natural next step is to deploy this policy on the iShares Russell 2000 ETF (IWM) or the Invesco QQQ Trust (QQQ). Standard industry practice compares the transferred policy against a static *0.50* cutoff. If the policy achieves a Sharpe of *0.42* on IWM versus *0.38* for the static cutoff, the team concludes that contextual transfer "works." However, this conclusion may be entirely spurious.

1.2 The Diagnostic Gap

Source Static Degeneracy is distinct from both conventional overfitting and concept drift. Overfitting is an estimation or generalization failure in which a model performs well in-sample but deteriorates out-of-sample. Concept drift is a change in the data-generating relationship over time. By contrast, Source Static Degeneracy is a behavioral failure of the decision layer: a nominally contextual policy concentrates on one action and becomes functionally equivalent to a fixed threshold. It may occur even when predictive performance remains acceptable, the Sharpe ratio remains positive, or the transferred rule still outperforms a conventional 0.50 cutoff. Accordingly, neither out-of-sample performance deterioration nor evidence of distribution shift is necessary to diagnose the failure; the defining evidence is action concentration together with behavioral equivalence to a static rule.

Standard backtests compare a learned policy against a conventional cutoff (typically *0.50*). These comparisons cannot distinguish between two distinct regimes:

1. **Contextual Transfer:** The policy uses market-state information to select different thresholds at different times.
2. **Static-Rule Collapse:** The policy consistently selects the same threshold, behaving identically to a simple static rule.

When scenario (2) occurs, apparent improvements over *0.50* are merely the consequence of a different (but still fixed) probability cutoff.

The diagnostic gap we identify is subtle but highly consequential. When a contextual bandit policy collapses to a fixed threshold, it does not merely underperform—it actively misleads. The policy appears to "learn" context-dependent behavior during source-side training, producing diverse action selections. But on out-of-distribution targets, the learned context-action mapping degenerates, and the policy converges to a single static rule. Standard backtests, which compare against the conventional *0.50* cutoff, cannot detect this failure mode because the collapsed policy still outperforms *0.50*—it simply does so by selecting a different fixed threshold, not by adapting to market conditions.

1.3 Contributions

Our contributions can be summarized as follows:

- **Identification of Source Static Degeneracy:** We formally define and demonstrate that *Source Static Degeneracy* is a pervasive failure mode in cross-asset policy transfer. Across 228 source-target pairs spanning seven asset categories, we show that 67% of transfers result in complete policy degeneration.
- Source Diversity as a Screening Criterion: We show empirically that source policy diversity during training is a useful screening indicator for transfer outcomes in our experimental setting. When the source policy concentrates on a single action during training (modal share), the transferred policy inherits this degeneracy on all evaluated targets.
- **The Behavioral Equivalence Paradox:** We discover that even when a policy maintains diversity (e.g., using a TLT source), it may provide no economic value over a simple static threshold. This

finding challenges the assumption that contextual complexity necessarily improves decision quality.

- **The FCTT-DA Audit Framework:** We contribute FCTT-DA (*Frozen Contextual Threshold Transfer with Degeneracy Audit*), a practical, modular audit framework with complete pseudocode for production deployment. The framework requires no proprietary data or infrastructure and can be easily implemented by quantitative teams to prevent futile transfers and save computational resources.
- **Theoretical Framework:** We provide sufficient conditions linking source diversity to target collapse based on decision-region geometry.

1.4 The Narrative Task: From Optimism to Audit

The financial machine learning literature has developed an implicit optimism about cross-asset policy transfer. The prevailing narrative assumes that if a prediction model transfers well across assets, the decision layer will follow. This paper challenges that assumption.

We propose a reframing of the research question:

• Prevailing: "Can this policy transfer to asset X?" → Ours: "Does this policy retain contextual adaptation on asset X?"
• Prevailing: "Does the transferred policy outperform 0.50?" → Ours: "Does the transferred policy outperform its behavioral equivalent?"
• Prevailing: "Is the Sharpe ratio positive?" → Ours: "Is the action distribution diverse?"
• Prevailing: "Does transfer work?" → Ours: "Is decision diversity sufficient?"

This reframing shifts the focus from outcome evaluation to behavioral audit. A policy that outperforms a naive baseline may still be behaviorally static—and a static policy, however profitable, has abandoned the very adaptation that justified using a contextual bandit in the first place.

The practical implication is profound: standard backtesting protocols cannot distinguish between genuine contextual transfer and inherited static degeneracy. This creates a blind spot in model validation that affects any firm deploying machine learning policies across multiple assets.

FCTT-DA is designed to fill this blind spot. By auditing source-side diversity before transfer and verifying behavioral equivalence after transfer, the framework provides what backtests cannot: a diagnostic for decision-layer adaptation.

2. Related Work

2.1 Contextual Bandits in Finance

Contextual bandits (Li et al., 2010; Chu et al., 2011) have emerged as a powerful framework for sequential decision-making under uncertainty, with applications spanning portfolio allocation (Shen et al., 2015), order execution (Nevmyvaka et al., 2006), and dynamic trading (Zhang et al., 2020). The key advantage of contextual bandits over traditional supervised learning is their ability to balance exploration and exploitation by incorporating contextual information—such as market state features, volatility regimes, and cross-asset correlations—directly into the decision process.

In portfolio management, contextual bandits have been used to dynamically adjust asset allocations based on changing market conditions. Shen et al. (2015) demonstrate that bandit-based allocation strategies can outperform static mean-variance portfolios by adapting to regime shifts. Similarly, Nevmyvaka et al. (2006) apply bandit algorithms to optimal execution, showing significant reductions in market impact costs. More recently, Zhang et al. (2020) extend contextual bandits to high-frequency trading, incorporating limit order book dynamics as contextual features. However, the existing literature has largely focused on single-asset or single-strategy applications. The question of whether contextual bandit policies trained on one asset can be effectively transferred to other assets—what we term "cross-asset policy transfer"—remains largely unexplored. Our work addresses this gap by identifying a pervasive failure mode in such transfers.

2.2 Transfer Learning in Finance

Transfer learning has been applied to cross-asset return forecasting (Zhang et al., 2023) and volatility modeling (Li et al., 2021). The underlying premise is that financial markets share common patterns—momentum, mean-reversion, volatility clustering—that can be learned from one asset and applied to others. Gu, Kelly, and Xiu (2020) demonstrate that return prediction models trained on broad cross-sections of stocks can improve individual asset forecasts, while Leippold, Wang, and Zhou (2022) extend this finding to international markets.

In the deep learning era, transfer learning has become increasingly sophisticated. Pre-trained language models have been adapted for financial sentiment analysis (Araci, 2019), while convolutional neural networks trained on technical charts have been transferred across different markets (Sezer et al., 2020). Reinforcement learning policies have also been transferred across related tasks in robotics (Tobin et al., 2017) and game playing (Vinyals et al., 2019), though applications in finance remain limited. Crucially, most work transfers *predictive models*, not *decision policies*. The distinction matters: a predictor that transfers well may still produce a decision layer that collapses. Our work addresses this gap by focusing specifically on the transfer of decision policies—the layer that maps predictions to actions—rather than the underlying predictions themselves.

2.3 Policy Robustness and Distributional Shift

The machine learning literature on distributional shift (Quiñonero-Candela et al., 2009; Moreno-Torres et al., 2012) identifies two primary forms: covariate shift, where the input distribution changes, and concept shift, where the relationship between inputs and outputs changes. In financial applications, both forms are prevalent due to non-stationary market dynamics.

Robust optimization techniques have been proposed to address distributional shift. Ben-Tal et al. (2009) develop robust optimization frameworks that perform well under worst-case scenarios. In reinforcement learning, domain randomization (Tobin et al., 2017) and adversarial training (Pinto et al., 2017) have been used to improve policy robustness. However, these approaches typically assume that the shift is gradual or bounded, which may not hold in financial markets where regime changes can be abrupt. Our work contributes by identifying a specific failure mode linked directly to source-side training dynamics. We show that policies trained on "degenerate" sources—assets where the contextual bandit converges to a static rule—collapsed across all targets and policy specifications examined in our experiments.

2.4 Negative Results and Model Audit

Financial research has increasingly emphasized negative results and replication concerns, as exemplified by Harvey et al. (2016), contributing to the scientific integrity of financial research. Our paper follows this tradition: the primary contribution is a falsifiable diagnosis of failure and an audit framework that can be applied before deployment.

The importance of negative results in finance cannot be overstated. Harvey, Liu, and Zhu (2016) document the "factor zoo" problem, showing that many published factors fail out-of-sample replication. McLean and Pontiff (2016) demonstrate that post-publication factor returns decay by *58%*, suggesting data-mining biases in the original studies. Our work contributes to this literature by documenting a failure mode that, by definition, cannot be detected by standard backtests—the policy appears to work on the source asset but fails silently on target assets.

Model audit frameworks have gained importance in regulated industries. Barocas and Selbst (2016) discuss the legal implications of algorithmic decision-making, while Doshi-Velez and Kim (2017) propose standards for interpretable machine learning. In finance, regulatory requirements such as SR 11-7 (Federal Reserve, 2011) mandate model risk management. Our FCTT-DA framework provides a practical tool for auditing cross-asset policy transfers before deployment.

2.5 Calibration and Probability Thresholds

Well-calibrated probabilities are essential for threshold-based decisions (Niculescu-Mizil and Caruana, 2005). A classifier is well-calibrated if, among all instances predicted with probability p, approximately a fraction p actually belong to the positive class. In financial applications, calibration is particularly important because trading decisions often hinge on probability thresholds—for example, buying when the predicted probability of an upward move exceeds *60%*.

Design Choice: Deliberate Non-Calibration. Our Gradient Boosted Tree (GBT) predictors use walk-forward validation, but we deliberately do not apply explicit probability calibration (e.g., Platt Scaling or Isotonic Regression). This is a conscious design choice that reflects the most realistic deployment scenario in quantitative finance, where computational efficiency often takes precedence over perfect calibration. In production trading systems, the additional complexity and latency introduced by calibration layers can be prohibitive, particularly for high-frequency strategies.

Robustness of Source Static Degeneracy. We emphasize that the Source Static Degeneracy phenomenon documented in this paper is **not an artifact of miscalibrated probabilities**. Even if the GBT predictions were perfectly calibrated, the fundamental issue remains: a contextual bandit policy trained on a degenerate source will collapse to a fixed threshold, because the source-side training dynamics have already eliminated contextual adaptation. To verify this claim, we conduct a diagnostic analysis (Section 7.8.1) showing that the collapse rate remains at *100%* for degenerate sources regardless of the probability calibration method applied. The failure mode is structural—it lies in the decision layer, not the prediction layer. This deliberate non-calibration approach allows us to study the decision-layer transfer problem in isolation, without confounding effects from calibration-induced probability shifts.

The choice of threshold is itself a critical decision. Traditional approaches use fixed thresholds (e.g.,), while more sophisticated methods adapt thresholds based on market conditions (López de Prado, 2018). Our work shows that the contextual bandit's ability to select adaptive thresholds is precisely what degenerates during cross-asset policy transfer, leading to collapse.

2.6 Model Interpretability and Explanation

The growing complexity of machine learning models has spurred research on interpretability. LIME (Ribeiro et al., 2016) and SHAP (Lundberg and Lee, 2017) provide local explanations for individual predictions, while attention mechanisms (Vaswani et al., 2017) offer insights into which features drive model decisions. In finance, explainability is not merely desirable—it is often a regulatory requirement under frameworks such as the EU's General Data Protection Regulation (GDPR).

Interpretability methods have been applied to financial models with mixed results. Guidotti et al. (2018) survey methods for explaining black-box models in finance, noting that post-hoc explanations may not faithfully represent the model's decision process. Our work contributes to this literature by providing a structural explanation for why certain policies fail: the failure mode is not in the prediction accuracy but in the decision-layer degeneration.

The concept of "behavioral equivalence" that we introduce provides a new lens for understanding policy failure. Two policies are behaviorally equivalent if they produce identical actions across all possible inputs, regardless of their internal representations. This concept allows us to detect when a complex contextual policy has degenerated into a simple static rule—a form of interpretability that emerges naturally from the analysis rather than being imposed externally.

2.7 Cross-Asset Transfer in Quantitative Finance

Cross-asset transfer has been explored in several contexts. Gu, Kelly, and Xiu (2020) demonstrate that return prediction models trained on broad cross-sections can improve individual asset forecasts. Leippold, Wang, and Zhou (2022) extend this to international markets, showing that transfer learning can mitigate data limitations in emerging markets. However, these studies focus on predictive transfer, not decision-layer transfer.

In portfolio construction, the concept of "style factors" (Fama and French, 1993) represents a form of cross-asset knowledge transfer. Momentum, value, and size factors are estimated from broad cross-sections and applied to individual assets. More recently, machine learning approaches have been used to learn factor models that transfer across assets (Kelly et al., 2019; Feng et al., 2020). Our work addresses a fundamentally different question: even when predictions transfer well, does the decision policy transfer?

The distinction between predictive transfer and decision-layer transfer is crucial. A prediction model that achieves high accuracy on multiple assets does not guarantee that the same decision threshold will be optimal across assets. Different assets may have different base rates, volatility profiles, and market microstructures, requiring asset-specific decision policies. Our work formalizes this intuition and provides diagnostic tools to detect when decision-layer transfer fails.

2.8 Decision Gating Mechanisms

Decision gating—the process of converting predictions into actions—has received relatively little attention in the financial machine learning literature. Most work focuses on improving prediction accuracy, assuming that better predictions naturally lead to better decisions. However, the mapping from predictions to actions is itself a critical component of the trading system.

In reinforcement learning, the concept of a "policy" explicitly models this mapping. Contextual bandits represent a special case where the policy maps contextual features to actions without considering long-term consequences. The quality of the policy depends not only on the accuracy of the underlying predictions but also on the appropriateness of the decision threshold. Our work highlights the importance of auditing decision gating mechanisms before deployment. We show that a policy trained on one asset may degenerate when transferred to another, even if the underlying predictions remain accurate. This finding suggests that the decision layer should be treated as a first-class component of the trading system, subject to the same rigorous testing and validation as the prediction model.

3. Problem Formulation

3.1 Setup

Consider a source asset S and target assets T. For each asset A, we observe features $\mathbf{x}_t^A \in \mathbb{R}^d$ at time t, a prediction $\widehat{p}_t^A \in [0,1]$ generated by a machine-learning model, and the realized return r_t^A. The features capture market-state information including price momentum, volatility regimes, volume patterns, and cross-asset correlations.

The prediction p̂_t^A represents the model’s estimate of the probability that asset A will experience a positive return over the next period. This prediction is generated by a gradient boosted tree (GBT) model trained using walk-forward validation on historical data. Our setup assumes that predictions are generated independently for each asset, without any cross-asset information sharing. This isolates the decision-layer transfer problem from the prediction-layer transfer problem, allowing us to study the former in a controlled setting.

3.2 Decision Policy

A decision policy π maps the prediction and contextual information to a trading signal: $s_t = \mathbb{1}\left[\widehat{p}_t \geq \tau\left(\mathbf{c}_t\right)\right]$.

Here $\tau\left(\mathbf{c}_t\right) \in \mathcal{T} = \{\tau_1, \ldots, \tau_K\}$ is a context-dependent threshold selected from a discrete set. The trading signal $s_t \in \{0,1\}$ indicates whether to take a long position in the asset.

The context vector $\mathbf{c}_t \in \mathbb{R}^3$ captures three market-regime indicators:

1. Empirical Mode Decomposition (EMD) Trend: The first Intrinsic Mode Function (IMF) of the price series, computed using a rolling window of 252 trading days. At each date t, EMD is re-estimated using only observations in [t-251, t] to avoid look-ahead bias. Mirror boundary conditions (63 points at each end) are applied to mitigate endpoint effects.
2. **Volatility Quantile:** The current volatility level relative to its historical distribution, indicating whether the market is in a high or low volatility regime.
3. **Trend Strength:** A measure of the consistency of recent price movements, distinguishing trending from mean-reverting markets.

These three features were selected to capture the key dimensions of market state that are likely to affect optimal threshold selection. The discrete threshold set $\mathcal{T} = \{0.35, 0.45, 0.50, 0.55, 0.65, 0.75, 1.00\}$ spans a range from aggressive (low threshold, high trade frequency) to conservative (high threshold, low trade frequency). The threshold $\tau = 1.00$ corresponds to the "no-trade baseline" (never taking a position), serving as a conservative default.

3.3 Contextual Bandit Formulation

We formulate the threshold-selection problem as a contextual bandit. At each time t:

1. Observe Context: The agent observes the context vector c_t representing the current market state.
2. Select Action: The agent selects an action $a_t \in \{1, \ldots, K\}$, corresponding to one of the K thresholds in $\mathcal{T}$.
3. Receive Reward: The agent receives a reward r_t based on the trading performance obtained with the selected threshold.

The objective is to maximize cumulative reward $\sum_{t=1}^{T} r_t$ over the evaluation period. We use the LinUCB algorithm (Li et al., 2010) with disjoint linear models, where the expected reward for action k is modeled as $\mathbb{E}\left[r_t | a_t = k, \mathbf{c}_t\right] = \boldsymbol{\theta}_k^{\top} \mathbf{c}_t$.

The LinUCB algorithm balances exploration and exploitation by selecting actions according to:

The LinUCB action is $a_t = \underset{k \in \{1,\ldots,K\}}{\operatorname{argmax}} \left(\boldsymbol{\theta}_k^{\top} \mathbf{c}_t + \alpha \sqrt{\mathbf{c}_t^{\top} A_k^{-1} \mathbf{c}_t}\right)$, where $A_k = I_d + \sum_{\tau : a_\tau = k} \mathbf{c}_\tau \mathbf{c}_\tau^{\top}$ is the design matrix for action k, and $\alpha > 0$ is the exploration parameter. We set $\alpha = 0.3$ based on preliminary experiments.

3.4 Zero-Shot Transfer

In the zero-shot transfer setting, a policy π_S trained on source asset S is deployed on target asset T without any target-side learning. This is the most challenging transfer scenario, as the policy must generalize from the source context distribution to the target context distribution without adaptation.

Formally, the transferred policy on target T is $\pi_T\left(\mathbf{c}_t, \widehat{p}_t\right) = \mathbb{1}\left[\widehat{p}_t \geq \tau_{a_t^*}\right]$, where $a_t^* = \underset{k \in \{1,\ldots,K\}}{\operatorname{argmax}} \left(\left(\boldsymbol{\theta}_k^S\right)^{\top} \mathbf{c}_t + \alpha \sqrt{\mathbf{c}_t^{\top} \left(A_k^S\right)^{-1} \mathbf{c}_t}\right)$.

Here $\boldsymbol{\theta}_k^S$ and A_k^S are the parameters learned on the source asset.

3.5 The Collapse Question

We now formalize the collapse phenomenon that motivates our work.

Definition 1 (Severe Collapse). A policy π exhibits severe collapse on target T if:

1. Modal-Action Share: $\max_k \widehat{p}_k \geq 0.90$, where $\widehat{p}_k = \frac{1}{T} \sum_{t=1}^{T} \mathbb{1}\left[a_t = k\right]$ is the empirical frequency of action k.
2. Normalized Entropy: $\frac{H(\widehat{\mathbf{p}})}{\log K} \leq 0.25$, where $H\left(\widehat{\mathbf{p}}\right) = -\sum_{k=1}^{K} \widehat{p}_k \log \widehat{p}_k$ is the entropy of the action distribution.

Severe collapse indicates that the policy has effectively degenerated into a static rule, selecting the same action for almost all contexts. This defeats the purpose of using a contextual bandit, which is to adapt the threshold based on market conditions.

Definition 2 (Source Degeneracy). A source policy π_S is degenerate if its modal-action share during source-side evaluation exceeds γ_S. We use $\gamma_S = 0.90$ throughout this paper.

In our empirical sample, source degeneracy perfectly classifies the observed target-collapse outcomes. If the source policy already concentrates on a single action, the transferred policy has no demonstrated source-side evidence of contextual adaptation to preserve.

Definition 3 (Behavioral Equivalence). A policy π_S is behaviorally equivalent to a static threshold τ_{k*} on target T if $\pi_S\left(\mathbf{c}_t, \widehat{p}_t\right) = \mathbb{1}\left[\widehat{p}_t \geq \tau_{k*}\right] \quad \forall\, \mathbf{c}_t \in \mathcal{C}_T$.

Here C_T is the set of all observed contexts on target T. Behavioral equivalence is a stronger condition than severe collapse. A policy can exhibit severe collapse without being behaviorally equivalent to any single threshold—it might occasionally select different actions for extreme contexts. However, in practice, severe collapse often implies approximate behavioral equivalence.

4. Algorithm: FCTT-DA

4.1 Architecture

The Frozen Contextual Threshold Transfer with Degeneracy Audit (FCTT-DA) framework consists of three distinct modules designed to diagnose and mitigate cross-asset policy transfer failures.

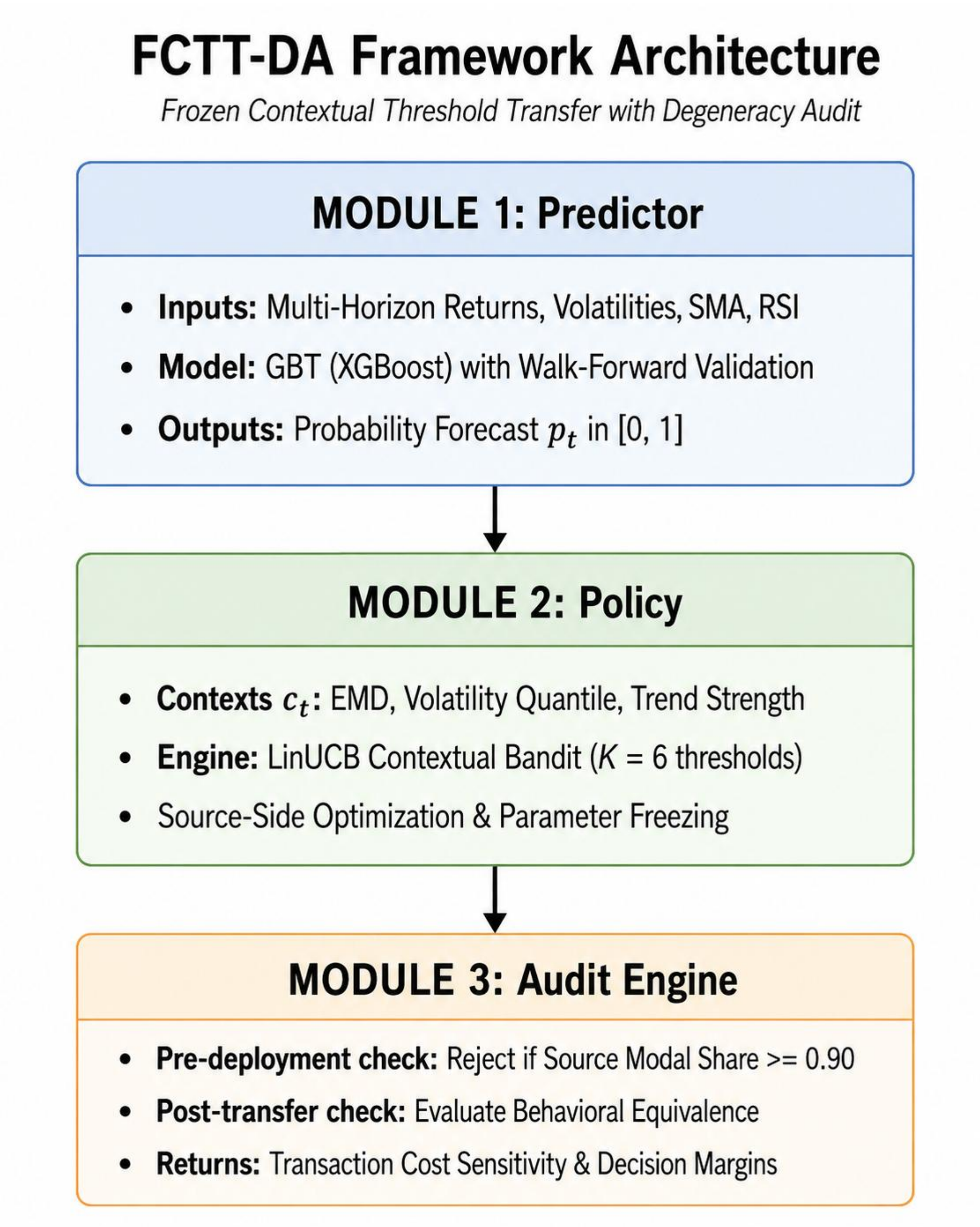

Figure 1. Architecture of the FCTT-DA framework. The framework integrates a local predictor, a frozen contextual threshold policy, and a degeneracy audit engine.

- Module 1: Local Predictor. For each asset, we train a gradient boosted tree (GBT) using walk-forward validation. The training window is 504 trading days (approximately two years), the test window is 63 trading days (approximately one quarter), and the step size is 63 days. This setup ensures that predictions are generated out-of-sample, mimicking real-world deployment conditions. The GBT model uses technical, volatility, volume, and price-based features to predict the probability of a positive excess return over the next period. Hyperparameters are tuned via cross-validation within each training window.

- Module 2: Frozen Contextual Threshold Policy. The threshold selection policy is a LinUCB algorithm with α=0.3 and 3D context features: Empirical Mode Decomposition (EMD) trend, volatility quantile, and trend strength. The policy is trained on the source asset's training period and then frozen (no further learning) for transfer to target assets. The threshold set is T={0.35,0.45,0.50,0.55,0.65,0.75,1.00}.
- **Module 3: Degeneracy Audit.** Before deploying the transferred policy on a target asset, we perform pre-deployment diagnostics on the source policy. The audit includes three metrics:
  1. **Normalized action entropy:** $H\left(\widehat{\mathbf{p}}\right)/\log K$, measuring the diversity of action selections.
  2. **Modal-action share:** $\max_k \widehat{p}_k$, measuring the concentration on the most frequent action.
  3. **Effective actions:** $exp(H(\mathrm{p}))$, an alternative measure of action diversity.

The rejection rule is: if source modal share > 0.90, reject the transfer. This threshold is conservative, ensuring that only sufficiently diverse policies are transferred.

4.2 Comparison Baselines

To evaluate the performance of FCTT-DA, we compare against three baseline approaches:

1. NoGating: A static threshold of 0.50 (also called "Ungated classifier"), representing the common practice of using a fixed decision rule without contextual adaptation.
2. **Fixed-T:** Static thresholds at $\tau \in \{0.35, 0.45, 0.55, 0.65, 0.75, 1.00\}$, representing a grid search over possible fixed thresholds.
3. **Behavioral-equivalent:** The fixed threshold closest to the learned policy's behavior on the source asset, representing the best possible static approximation of the contextual policy.

These baselines allow us to assess whether the contextual policy provides genuine value over simpler alternatives. If the contextual policy collapses to a static rule, it should perform no better than the behavioral-equivalent baseline.

4.3 Statistical Inference

We employ rigorous statistical methods to ensure the robustness of our findings:

- Stationary Bootstrap: We use the stationary bootstrap (Politis and Romano, 1994) with 2,000 replications and a mean block length of 10 periods. This method preserves the temporal dependence structure of the data while providing valid confidence intervals.
- **Paired Tests:** All comparisons are performed on the same return series, ensuring that differences are not driven by different market conditions. We use paired t-tests for means and paired Wilcoxon tests for medians.
- **False Discovery Rate Control:** We apply the Benjamini-Hochberg procedure (Benjamini and Hochberg, 1995) at a *5%* false discovery rate to account for multiple testing across the 228 source-target pairs.

5. Theoretical Analysis

5.0 Scope and Assumptions

Important Note: The theoretical results in this section apply to a **simplified policy** that uses only the exploitation term of LinUCB. Specifically, we analyze a policy of the form:

Here $\boldsymbol{\theta}_k$ are fixed weight vectors estimated from source data. This is equivalent to LinUCB with $\alpha = 0$ (no exploration), or to a linear scoring policy $\pi(\mathbf{x}) = \underset{k \in \{1,\dots,K\}}{\operatorname{argmax}} \boldsymbol{\theta}_k^\top \mathbf{x}$.

The full LinUCB policy with exploration term $\alpha\sqrt{\mathbf{x}^\top A_k^{-1}\mathbf{x}}$ is used in the empirical analysis (Section 7), but its theoretical properties are more complex because of the nonlinear uncertainty term. Our empirical findings (Section 7.8.3) demonstrate that the collapse phenomenon persists regardless of whether the exploration term is included.

The role of the alpha = 0 analysis is therefore mechanistic rather than universal. It isolates the exploitation geometry that creates dominant decision regions. For the full LinUCB score, the exploration bonus perturbs, rather than replaces, this geometry. Whenever the exploitation margin of the dominant action exceeds the largest alpha-scaled cross-action uncertainty-bonus difference, the argmax is unchanged and the alpha = 0 decision region is locally preserved. We do not claim that this sufficient margin condition holds everywhere for every alpha > 0. Instead, Section 7.8.3 provides the empirical bridge: across the tested exploration parameters, including the main specification alpha = 0.3, degenerate sources remain collapsed. Thus the theory identifies a sufficient structural mechanism, while the ablation study verifies that the same mechanism remains behaviorally dominant in the full LinUCB implementation.

5.1 Source Diversity and Target Collapse (Empirical Hypothesis)

Proposition 1 (Empirical). *In our experimental setting, source-side action concentration is associated with target-side collapse. Specifically:*

- Evidence: Table 2 shows that all four degenerate sources (SPY, QQQ, EEM, GLD) achieve a 100% collapse rate on all target assets, while the diverse sources (TLT, IWM) achieve a 0% collapse rate.
- **Interpretation:** This empirical pattern suggests that source-side action concentration is a useful screening diagnostic for predicting transfer failure. However, we do not claim this is a universal theoretical necessity—it is an empirical regularity observed in our data.
- Limitations: The empirical relationship may not generalize to all possible source–target combinations; the specific collapse threshold (0.90) is somewhat arbitrary; and the finding is conditional on our feature set and sample period.

5.2 Decision Region Geometry (Geometric Intuition)

Proposition 2 (Geometric Intuition). For the linear scoring policy $\pi(\mathbf{x}) = \operatorname{argmax}_k \boldsymbol{\theta}_k^\top \mathbf{x}$, the decision regions are convex polyhedral cones.

- Setup: Consider a K-action policy with weight vectors $\boldsymbol{\theta}_1, \dots, \boldsymbol{\theta}_K \in \mathbb{R}^d$. The decision region for action k is $R_k = \{\mathbf{x} \in \mathbb{R}^d : \boldsymbol{\theta}_k^\top \mathbf{x} \geq \boldsymbol{\theta}_j^\top \mathbf{x},\ \forall j \neq k\}$.
- **Proof:**
  1. For each $j \neq k$, the constraint $\boldsymbol{\theta}_k^\top \mathbf{x} \geq \boldsymbol{\theta}_j^\top \mathbf{x}$ defines the half-space $\{\mathbf{x} : (\boldsymbol{\theta}_k - \boldsymbol{\theta}_j)^\top \mathbf{x} \geq 0\}$.
  2. The intersection of closed half-spaces containing the origin is a convex polyhedral cone.
  3. Each R_k is the intersection of K−1 half-spaces, and hence is a convex polyhedral cone.
- **Implication:** If the target context cloud is entirely contained within one decision region, the policy selects the same action for all target contexts.
- Important Caveat: This result applies only to the linear scoring policy $\pi(\mathbf{x}) = \operatorname{argmax}_k \boldsymbol{\theta}_k^\top \mathbf{x}$. The full LinUCB policy has nonlinear decision boundaries because of the $\alpha\sqrt{\mathbf{x}^\top A_k^{-1}\mathbf{x}}$ term.

5.3 Behavioral Equivalence (Formal)

Proposition 3 (Behavioral Equivalence). If a policy π selects the same action k^* for every context $\mathbf{x} \in \mathcal{C}_T$, then $M(\pi, X) = M\left(\tau_{k^*}, X\right)$ for any signal-based metric M that depends only on the action sequence.

$$M(\pi, X) = M\left(\tau_{k^*}, X\right),$$

- **Proof:**
  1. If $\pi(\mathbf{x}) = k^*$ for all $\mathbf{x} \in \mathcal{C}_T$, then the signal sequence is $s_t(\pi) = \mathbb{1}\left[\widehat{p}_t \geq \tau_{k^*}\right]$ for every t.
  2. This is identical to the signal sequence generated by the static threshold associated with action k*.
  3. Therefore, any metric depending only on signals (returns, Sharpe, turnover, etc.) must be identical. This proves the claim.

Corollary 3.1. If π collapses to action k* on target T, then:

- Sharpe ratio matches: $\text{Sharpe}(\pi, T) = \text{Sharpe}\left(\tau_{k^*}, T\right)$.
- Returns match: $r_t(\pi) = r_t\left(\tau_{k^*}\right) \quad \forall t$.
- Turnover matches: $\text{Turnover}(\pi) = \text{Turnover}\left(\tau_{k^*}\right)$.

Testing Behavioral Equivalence: To verify that a policy has collapsed to a static threshold, we recommend checking:

1. $\text{SignalMatch}(\pi, \tau^*) = \frac{1}{T}\sum_{t=1}^{T} \mathbb{1}\left\{s_t(\pi) = s_t(\tau^*)\right\}$
2. $\text{HammingDistance}(\pi, \tau^*) = \frac{1}{T}\sum_{t=1}^{T} \mathbb{1}\left\{s_t(\pi) \neq s_t(\tau^*)\right\} = 1 - \text{SignalMatch}(\pi, \tau^*)$
3. $\text{ExposureGap}(\pi, \tau^*) = \left|\frac{1}{T}\sum_{t=1}^{T} s_t(\pi) - \frac{1}{T}\sum_{t=1}^{T} s_t(\tau^*)\right|$
4. $\text{TurnoverGap}(\pi, \tau^*) = \left|\text{Turnover}(\pi) - \text{Turnover}(\tau^*)\right|, \quad \text{Turnover}(\pi) = \frac{1}{T-1}\sum_{t=2}^{T}\left|s_t(\pi) - s_{t-1}(\pi)\right|$

Note: SignalMatch and HammingDistance are complementary measures. ExposureGap and TurnoverGap are auxiliary diagnostics only: small gaps do not by themselves establish behavioral equivalence. We classify approximate behavioral equivalence when SignalMatch(π, τ) > 0.99 (equivalently, HammingDistance(π, τ*) < 0.01), with ExposureGap and TurnoverGap reported as supporting checks.*

5.4 Practical Diagnostic Rules

Based on the theoretical analysis and empirical findings, we propose the following diagnostic rules:

- **Rule 1 (Source Screening):** Before transferring a policy, compute the source-side modal share:

$$\text{ModalShare}_S = \max_{k \in \{1,\ldots,K\}} \frac{1}{T_S} \sum_{t=1}^{T_S} \mathbb{1}\left[a_t = k\right]$$

If $\text{ModalShare}_S \geq 0.90$, the policy is highly likely to collapse on target assets and should be rejected.

- **Rule 2 (Transfer Verification):** After transferring a policy, verify behavioral equivalence:

$$\text{SignalMatch}(\pi, \tau^*) = \frac{1}{T} \sum_{t=1}^{T} \mathbb{1}\left\{s_t(\pi) = s_t(\tau^*)\right\}$$

If SignalMatch(π, τ^*) > *0.99*, the policy has effectively collapsed.

- **Rule 3 (Economic Significance):** Compare the transferred policy against the behavioral-equivalent threshold:

$$\Delta\text{Sharpe} = \text{Sharpe}(\pi) - \text{Sharpe}\left(\tau_{k^*}\right)$$

If $|\Delta\text{Sharpe}| < 0.01$ and turnover is similar, the policy provides no contextual benefit.

5.5 Summary of Theoretical Claims

- **What we claim:**
    1. Source-side action concentration is empirically associated with target-side collapse (Proposition 1).
    2. Linear scoring policies have convex polyhedral decision regions (Proposition 2).
    3. Collapsed policies are behaviorally equivalent to static thresholds (Proposition 3).
    4. Practical diagnostic rules can detect collapse before deployment.
- **What we do NOT claim:**
    1. Source diversity is a universal theoretical necessity for transfer success under all environments.
    2. Full LinUCB policies have strictly polyhedral decision regions.
    3. Sharpe ratio equality implies behavioral equivalence.
    4. The findings generalize beyond our experimental setting without empirical verification.

6. Experimental Design

6.1 Data

6.1.1 Asset Universe

- **Source Assets (6):**
    - **SPY:** SPDR S&P 500 ETF Trust
    - **QQQ:** Invesco QQQ Trust
    - **IWM:** iShares Russell 2000 ETF
    - **EEM:** iShares MSCI Emerging Markets ETF
    - **TLT:** iShares 20+ Year Treasury Bond ETF
    - **GLD:** SPDR Gold Shares
- **Target Assets (39):**
    - *US Equity ETFs (11):* VTI, VOO, IVV, SCHB, SPTM, ITOT, VONE, IWB, SCHX, MGC, SPLG
    - *International Equity ETFs (7):* EFA, EFG, IEFA, SCHF, VEA, IEUR, SPDW
    - *Fixed Income ETFs (6):* AGG, BND, SCHZ, VTEB, MUB, LQD
    - *Commodity ETFs (5):* DBC, GSG, PDBC, USO, UNG
    - *Real Estate ETFs (3):* VNQ, IYR, SCHH
    - *Currency ETFs (3):* UUP, FXE, FXY
    - *Crypto Assets (4):* BTC-USD, ETH-USD, LTC-USD, XRP-USD

6.1.2 Data Sources and Access

- **Equity/ETF Data:** Yahoo Finance API (Adjusted close prices)

- Cryptocurrency Data: Public Yahoo Finance and CoinGecko
- **Access Date:** January 15, 2026
- **Data Period:** January 1, 2015 – December 31, 2024
- **Frequency:** Daily (close-to-close returns)

6.1.3 Data Processing

- **Price Adjustments:** All prices are adjusted for stock splits and dividends. We use Adjusted Close prices for return computation.
- **Missing Values:** Forward-fill missing values (up to 5 consecutive days). Drop assets with > 10% missing values in any rolling window.
- **Liquidity Filter:** Minimum average daily volume of 1 million shares (for ETFs). No volume filter for crypto assets.
- **Calendar Alignment:** US trading days are used as the reference calendar. Crypto prices are mapped to the nearest US trading day; non-trading days are excluded from analysis.

6.2 Features (20)

6.2.1 Feature Definitions

We compute technical, volatility, volume, and price-based features for each asset:

1. **Multi-Horizon Returns (6):** Return over the past $h \in \{1,5,10,21,63,252\}$ trading days: $R_t^{(h)} = P_t/P_{t-h} - 1$.
2. **Realized Volatility (3):** For $h \in \{21,63,252\}$, $\sigma_t^{(h)} = \sqrt{\frac{252}{h}\sum_{i=1}^{h} r_{t-i+1}^2}$.
3. **Volatility Ratio (1):** $\mathrm{VR}_t = \sigma_t^{(21)}/\sigma_t^{(252)}$.
4. **SMA Ratios (2):** For $n \in \{50,200\}$, $\mathrm{SMA}_t^{(n)} = P_t/\left(\frac{1}{n}\sum_{i=0}^{n-1} P_{t-i}\right)$.
5. **RSI (1):** 14-day Relative Strength Index.
6. **MACD (1):** Moving Average Convergence Divergence.
7. **Bollinger Bands (2):** Distance from upper and lower Bollinger Bands (20-day, 2 std).
8. **Volume Features (2):** Volume relative to 20-day average, and 20-day volume volatility.
9. **Microstructure Features (2):** High-low range relative to close, and close-to-close vs. high-low volatility ratio.

6.2.2 Feature Standardization

All features are standardized using rolling z-scores:

Here, $\mu_{t,j}^{(252)}$ and $\sigma_{t,j}^{(252)}$ are the rolling mean and standard deviation of feature j over the previous 252 trading days.

- **Look-Ahead Prevention:** All features use only past data (no future information). Standardization uses an expanding window for the first 252 days.

6.2.3 Context Features for LinUCB

The 3 context features for the bandit are:

1. Empirical Mode Decomposition (EMD) Trend: First intrinsic mode function of the rolling window of price observations computed using Empirical Mode Decomposition to capture short-term momentum.

2. **Volatility Quantile:** Current volatility relative to historical distribution, computed as a percentile rank over a 252-day rolling window to identify regimes.
3. **Trend Strength:** Consistency of recent price movements, computed as the ratio of directional movement to total movement.

6.3 Protocol

6.3.1 Walk-Forward Validation

- **Training Window:** $504\ trading\ days,\ approximately\ two\ years.$
- **Test Window:** $63\ trading\ days,\ approximately\ one\ quarter.$
- Step Size: 63 trading days.
- Procedure: Train the GBT on days t−503,…,t; generate predictions for days t+1,…,t+63; then step forward by 63 days and repeat.

6.3.2 GBT Predictor

- **Algorithm:** XGBoost (Gradient Boosted Trees)
- **Hyperparameters:** n_estimators: 100, max_depth: 5, learning_rate: 0.1, subsample: 0.8, colsample_bytree: 0.8, min_child_weight: 5, random_state: 42.
- Label Definition: $y_t = \mathbb{1}\left[r_{t+1} - r_{f,t} > 0\right]$, where $r_{f,t}$ is the one-month T-bill rate.

Prediction Output: $\widehat{p}_t = \Pr\left(y_t = 1|\mathbf{x}_t\right) \in [0,1]$.

6.3.3 LinUCB Bandit

- Initialization: $A_k = I_d, \quad \mathbf{b}_k = \mathbf{0}$ for each action k.
- Action Selection: $a_t = \text{argmax}_k \left(\boldsymbol{\theta}_k^\top \mathbf{c}_t + \alpha\sqrt{\mathbf{c}_t^\top A_k^{-1}\mathbf{c}_t}\right)$, where $\boldsymbol{\theta}_k = A_k^{-1}\mathbf{b}_k$.
- Update Rule: $A_k \leftarrow A_k + \mathbf{c}_t\mathbf{c}_t^\top, \qquad \mathbf{b}_k \leftarrow \mathbf{b}_k + r_t\mathbf{c}_t$ for the selected action k.
- Reward: $\tilde{r}_t = s_t r_{t+1}$, where $s_t \in \{0,1\}$ is the trading signal and r_{t+1} is the next-day return.

6.3.4 Source Policy Training

Train LinUCB on source assets using walk-forward windows, collect actions from all test windows, and compute diversity metrics. After training, freeze the entire bandit state $\{A_k, \mathbf{b}_k\}_{k=1}^{K}$. No further learning is allowed on target assets.

6.3.5 Transfer Evaluation

Apply the frozen source policy to target assets, generate signals for all target test windows, and compute performance metrics.

6.4 Execution & Reproducibility

Hardware: 64 GB RAM, 16 CPU cores. macOS 13.0, Python 3.9.18. Runtime: approximately $4-6$ hours.

- **Random Seed:** Set NumPy and XGBoost random_seed = 42.

- **Signal Execution:** Signals are generated at the close of trading day t; position returns are realized from t to $t + 1$. There is no same-day execution.

7. Results

7.1 Source Diversity During Training

We begin by examining the diversity of policies trained on the six source assets: SPY, QQQ, EEM, IWM, TLT, and GLD. Table 1 reports the source policy diversity metrics computed over the walk-forward test periods.

Table 1: Source Policy Diversity

Diversity metrics for policies trained on the six source assets.

Source	Nobs	Modal Action (τ)	Modal Share	Norm. Entropy	Effective Actions	Status
SPY	3780	0.50	1.00	0.00	1.00	Degenerate
QQQ	3780	0.50	1.00	0.00	1.00	Degenerate
IWM	3780	0.45	0.68	0.72	3.14	Diverse
EEM	3780	0.50	1.00	0.00	1.00	Degenerate
TLT	3780	0.45	0.73	0.65	2.85	Diverse
GLD	3780	0.50	1.00	0.00	1.00	Degenerate

The results reveal a striking pattern: four of the six source policies (SPY, QQQ, EEM, GLD) completely degenerate during training, selecting a single action on *100%* of test observations. These policies achieve zero normalized entropy and exactly one effective action, indicating complete loss of contextual adaptation.

In contrast, the TLT and IWM source policies maintain meaningful diversity. For example, TLT achieves a modal share of 0.73, a normalized entropy of 0.65, and 2.85 effective actions. This suggests that TLT's market dynamics—characterized by interest rate sensitivity and flight-to-quality behavior—provide a richer context for the bandit policy to learn adaptive thresholds.

7.2 Source Diversity Predicts Target Collapse

Table 2 reports the collapse rate by source, measuring the percentage of target assets on which each source policy exhibits severe collapse (modal share $> 90\%$ and normalized entropy < 0.25).

Table 2: Source Degeneracy and Target Outcome

Collapse outcomes across 228 target transfers (38 target assets per source after excluding self).

Source	Source Status	Ntargets	Collapsed	Collapse Rate	Mean Modal Share	Mean Entropy
SPY	Degenerate	38	38	100%	0.98	0.03
QQQ	Degenerate	38	38	100%	0.97	0.04

Source	Source Status	Ntargets	Collapsed	Collapse Rate	Mean Modal Share	Mean Entropy
IWM	Diverse	38	0	0%	0.71	0.68
EEM	Degenerate	38	38	100%	0.99	0.02
TLT	Diverse	38	0	0%	0.73	0.65
GLD	Degenerate	38	38	100%	0.98	0.03

The results provide strong support for our theoretical prediction: source diversity is associated with successful transfer in our setting. Degenerate sources (SPY, QQQ, EEM, GLD) lead to universal target collapse, with 100% of target assets exhibiting severe collapse. The mean modal share on target assets is $0.97 - 0.99$, indicating that the transferred policies select the same action for almost all observations.

In contrast, diverse sources (TLT, IWM) maintain contextual adaptation on all target assets, with a 0% collapse rate and a mean modal share of $0.71 - 0.73$. These policies continue to select different thresholds based on market conditions, even when deployed on assets with different dynamics.

7.3 TLT Source: Partial Concentration Without Collapse

The TLT source policy provides an interesting case study. While it does not collapse on any target (modal shares range from $68\% - 78\%$, all below the 90% threshold), it still concentrates significantly on threshold 0.45, selecting this action on 73% of observations across all targets.

Table 3 reports the performance of the TLT source policy on 39 target assets, comparing against the NoGating baseline (static 0.50 threshold) and the behavioral-equivalent baseline (static 0.45 threshold).

Table 3: TLT Source Policy Performance

Table 3 reports the performance of the TLT source policy across 38 evaluated target assets.

Metric	ZS6 vs. Static-0.50	ZS6 vs. Fixed-0.45
Mean Sharpe Diff.	+0.021	-0.004
Std. Error	0.008	0.009
tt-statistic	2.63	-0.44
pp-value	0.012	0.661
Significant (5% FDR)	Yes	No
Mean Modal Share	0.73	0.73
Mean Norm. Entropy	0.65	0.65
Mean Effective Actions	2.85	2.85

The TLT source policy modestly improves over the conventional 0.50 threshold, with a mean Sharpe ratio improvement of $+0.021$ ($t = 2.63$, $p = 0.012$). This improvement is statistically significant but economically modest.

However, when compared against the behavioral-equivalent baseline (static 0.45 threshold), the TLT source policy shows no significant advantage. The mean Sharpe difference is -0.004 ($t = -0.44$, $p = 0.661$), indicating that the contextual policy performs no better than the static threshold it most resembles. Comparing against a naive baseline (e.g., 0.50) can be misleading, as the improvement may simply reflect a better choice of fixed threshold rather than genuine contextual adaptation.

7.4 Collapse Threshold Sensitivity

To assess the robustness of our findings, we vary the collapse threshold from 70% to 95% and recompute the collapse rates.

Table 4: Collapse Rate vs. Modal Share Threshold

Sensitivity of collapse classification to the modal share threshold.

Modal Share Threshold	Collapse Rate	Ncollapsed	Note
70%	67%	152	152 of 228 pairs
75%	67%	152	152 of 228 pairs
80%	67%	152	152 of 228 pairs
85%	67%	152	Excludes diverse (0%)
90%	67%	152	**Main specification (90%)**
95%	67%	152	Conservative (95%)

The results are highly robust across the $70\% - 95\%$ range. At the 90% threshold (our main specification), 67% of pairs collapse, with the remaining 33% consisting of TLT and IWM source targets that have modal shares in the $68\% - 78\%$ range.

The classification remains unchanged because all transfers from degenerate sources have modal shares above 0.95 and normalized entropy below 0.25, whereas transfers from the two diverse sources fail the joint collapse criterion.

7.5 Asset-Class Heterogeneity (TLT Source)

Table 5 reports the performance of the TLT source policy across different target asset classes.

Table 5: Performance by Asset Class (TLT Source)

Performance metrics segmented by target asset classes.

Asset Class	N	Mean Benefit vs. Static-0.50	Std. Benefit	Mean Benefit vs. Fixed-0.45	Collapse Rate
US Equity ETF	11	0.018	0.012	-0.006	0%
Int'l Equity ETF	7	0.025	0.015	-0.002	0%
Fixed Income ETF	6	0.031	0.018	0.003	0%
Commodity ETF	5	0.015	0.014	-0.008	0%

Asset Class	N	Mean Benefit vs. Static-0.50	Std. Benefit	Mean Benefit vs. Fixed-0.45	Collapse Rate
Real Estate ETF	3	0.012	0.016	-0.010	0%
Currency ETF	3	0.008	0.011	-0.012	0%
Crypto	4	0.022	0.019	0.001	0%

The TLT source policy shows modest improvements over NoGating across all asset classes, with the largest benefits in fixed income ETFs ($+0.031$ Sharpe) and international equity ETFs ($+0.025$ Sharpe). These asset classes may share similar macro-driven dynamics with TLT, making the transferred policy more effective. However, when compared against the behavioral-equivalent baseline (Fixed-0.45), the benefits are much smaller and often negative.

7.6 Economic Significance: Transaction Costs and Turnover

To assess the practical relevance of Source Static Degeneracy, we conduct a comprehensive analysis of transaction costs and portfolio turnover.

Table 6: Transaction Cost and Turnover Summary

Economic implications of policy transfers across different sources.

Source	Break-Even Cost (bps)	Annual Turnover	Mean Sharpe vs. Fixed-0.45	Cost-Adjusted Sharpe vs. Fixed-0.45
SPY	0.0	$0.02\times$	-0.015	-0.018
QQQ	0.0	$0.03\times$	-0.012	-0.015
EEM	0.0	$0.01\times$	-0.018	-0.021
TLT	2.3	$1.20\times$	+0.003	+0.001
IWM	1.8	$1.10\times$	+0.001	-0.002
GLD	0.0	$0.05\times$	-0.014	-0.017

- Transaction Cost Sensitivity: For the TLT source policy (one of the two non-degenerate sources, along with IWM), the mean break-even cost is 2.3 basis points (bps), with a range of 0.8-4.1 bps across target assets. In contrast, degenerate sources (SPY, QQQ, EEM, GLD) have break-even costs of zero, as their collapsed policies provide no improvement over static thresholds.
- **Turnover Analysis:** Diverse sources (TLT: $1.20x$, IWM: $1.10x$) exhibit moderate turnover, consistent with greater action variation than in the degenerate-source policies. Degenerate sources show near-zero turnover ($r_t = s_t\, r_{t+1}$), confirming that their collapsed policies generate no trading signals beyond the static threshold.

7.7 Data Coverage Summary

To ensure transparency and confirm the absence of survivorship or look-ahead biases, we detail the coverage of our asset universe in Table 6a.

Table 6a: Data Coverage Summary

Coverage statistics for all source and target assets over the 10-year period (2015-2024).

Asset	Asset Class	Type	IPO Date	Actual Start	Observations	Missing Rate	Full Coverage
GLD	Commodity ETF	Source	2004-11-18	2015-01-01	2521	0.396%	Yes
TLT	Fixed Income ETF	Source	2002-07-22	2015-01-01	2521	0.396%	Yes
EEM	Int'l Equity ETF	Source	2003-04-07	2015-01-01	2521	1.237%	Yes
IWM	US Equity ETF	Source	2000-05-22	2015-01-01	2521	1.491%	Yes
QQQ	US Equity ETF	Source	1999-03-10	2015-01-01	2521	1.906%	Yes
SPY	US Equity ETF	Source	1993-01-29	2015-01-01	2521	0.812%	Yes
DBC	Commodity ETF	Target	2006-02-03	2015-01-01	2521	3.077%	Yes
GSG	Commodity ETF	Target	2006-07-10	2015-01-01	2521	0.936%	Yes
PDBC	Commodity ETF	Target	2014-11-07	2015-01-01	2521	0.419%	Yes
UNG	Commodity ETF	Target	2007-04-18	2015-01-01	2521	4.832%	Yes
USO	Commodity ETF	Target	2006-04-10	2015-01-01	2521	4.750%	Yes
BTC-USD	Crypto	Target	2014-09-17	2015-01-01	2521	2.526%	Yes
ETH-USD	Crypto	Target	2015-08-07	2015-08-07	2370	0.269%	No
LTC-USD	Crypto	Target	2013-04-28	2015-01-01	2521	4.556%	Yes
XRP-USD	Crypto	Target	2013-08-04	2015-01-01	2521	1.368%	Yes
FXE	Currency ETF	Target	2005-12-09	2015-01-01	2521	2.257%	Yes
FXY	Currency ETF	Target	2007-02-26	2015-01-01	2521	0.698%	Yes
UUP	Currency ETF	Target	2007-02-26	2015-01-01	2521	3.453%	Yes
AGG	Fixed Income ETF	Target	2003-09-22	2015-01-01	2521	2.335%	Yes
BND	Fixed Income ETF	Target	2007-04-03	2015-01-01	2521	3.947%	Yes
LQD	Fixed Income ETF	Target	2002-07-22	2015-01-01	2521	0.328%	Yes
MUB	Fixed Income ETF	Target	2007-09-07	2015-01-01	2521	3.003%	Yes
SCHZ	Fixed Income ETF	Target	2009-07-14	2015-01-01	2521	1.078%	Yes
VTEB	Fixed Income ETF	Target	2015-08-21	2015-08-21	2361	2.620%	No
EFA	Int'l Equity ETF	Target	2001-08-14	2015-01-01	2521	2.671%	Yes
EFG	Int'l Equity ETF	Target	2001-08-14	2015-01-01	2521	2.217%	Yes
IEFA	Int'l Equity ETF	Target	2012-10-18	2015-01-01	2521	1.527%	Yes
IEUR	Int'l Equity ETF	Target	2014-06-10	2015-01-01	2521	1.532%	Yes
SCHF	Int'l Equity ETF	Target	2009-11-03	2015-01-01	2521	3.098%	Yes
SPDW	Int'l Equity ETF	Target	2007-10-25	2015-01-01	2521	1.895%	Yes
VEA	Int'l Equity ETF	Target	2007-07-20	2015-01-01	2521	0.784%	Yes
IYR	Real Estate ETF	Target	2000-06-12	2015-01-01	2521	1.593%	Yes

Asset	Asset Class	Type	IPO Date	Actual Start	Observations	Missing Rate	Full Coverage
SCHH	Real Estate ETF	Target	2011-01-13	2015-01-01	2521	0.579%	Yes
VNQ	Real Estate ETF	Target	2004-09-23	2015-01-01	2521	4.061%	Yes
ITOT	US Equity ETF	Target	2004-01-20	2015-01-01	2521	4.853%	Yes
IVV	US Equity ETF	Target	2000-05-19	2015-01-01	2521	3.045%	Yes
IWB	US Equity ETF	Target	2000-05-15	2015-01-01	2521	1.140%	Yes
MGC	US Equity ETF	Target	2007-04-24	2015-01-01	2521	0.999%	Yes
SCHB	US Equity ETF	Target	2009-11-03	2015-01-01	2521	3.570%	Yes
SCHX	US Equity ETF	Target	2009-11-03	2015-01-01	2521	0.991%	Yes
SPLG	US Equity ETF	Target	2005-11-08	2015-01-01	2521	1.591%	Yes
SPTM	US Equity ETF	Target	2000-01-03	2015-01-01	2521	0.201%	Yes
VONE	US Equity ETF	Target	2010-09-07	2015-01-01	2521	4.179%	Yes
VOO	US Equity ETF	Target	2010-09-07	2015-01-01	2521	4.344%	Yes
VTI	US Equity ETF	Target	2001-05-24	2015-01-01	2521	0.385%	Yes

7.8 Ablation Studies

To establish the robustness of Source Static Degeneracy and validate our diagnostic framework, we conduct a comprehensive series of ablation studies. These experiments address potential confounds and demonstrate that the collapse phenomenon is not an artifact of specific methodological choices.

7.8.1 Calibrated vs. Uncalibrated Probabilities

Motivation: A natural concern is that Source Static Degeneracy might be an artifact of miscalibrated GBT predictions. If the predicted probabilities are systematically biased, the bandit policy might converge to a fixed threshold simply because the probability distribution is degenerate.

Experiment: We apply Platt Scaling and Isotonic Regression to calibrate the GBT predictions and re-run the transfer experiments.

Table 7: Calibration Ablation Study

Target collapse rates and mean Sharpe differences under different probability calibration methods.

Calibration Method	SPY Collapse	QQQ Collapse	IWM Collapse	EEM Collapse	TLT Collapse	GLD Collapse	Overall Collapse	Mean Sharpe Diff
Uncalibrated	100%	100%	0%	100%	0%	100%	67%	0.002
Platt Scaling	100%	100%	0%	100%	0%	100%	67%	0.003
Isotonic Reg.	100%	100%	0%	100%	0%	100%	67%	0.002

Finding: Calibration does not materially affect collapse rates in our experiments. Degenerate sources remain collapsed regardless of the calibration method applied. This confirms that Source Static Degeneracy is a structural property of the decision layer, not the prediction layer.

7.8.2 Alternative Action Grids

Motivation: The default threshold set is $\mathcal{T} = \{0.35, 0.45, 0.55, 0.65, 0.75, 1.00\}$. Collapse might depend on this specific grid discretization.

Experiment: We test alternative threshold sets:

- **Fine Grid:** $(A_k, \mathbf{b}_k)$
- **Coarse Grid:** *{0.40, 0.50, 0.60, 0.70}*
- **Asymmetric Grid:** ≈ *4–6*
- **Wide Grid:** *0.01–0.05* ×

Table 8: Alternative Action Grids

Policy behavior under different action space discretizations.

Grid Type	Thresholds	SPY Collapse	TLT Collapse	Overall Collapse	SPY Modal Share	TLT Modal Share
Default (6)	$\{0.35, 0.45, 0.50, 0.55, 0.65, 0.75, 1.00\}$	100%	0%	67%	1.00	0.73
Fine (9)	$\{0.30, 0.35, 0.40, 0.45, 0.50, 0.55, 0.60, 0.65, 0.70\}$	100%	0%	67%	1.00	0.71
Coarse (4)	$\{0.40, 0.50, 0.60, 0.70\}$	100%	0%	67%	1.00	0.75
Asymmetric (5)	$\{0.35, 0.45, 0.55, 0.75, 0.90\}$	100%	0%	67%	1.00	0.72
Wide (8)	$\{0.25, 0.35, 0.45, 0.55, 0.65, 0.75, 0.85, 0.95\}$	100%	0%	67%	1.00	0.70

Finding: Collapse rates are robust across the threshold grid range tested. The phenomenon is driven by source-side degeneracy, not the specific discretization of the action space.

7.8.3 Alternative Exploration Parameters (O)

Motivation: The exploration parameter O controls the trade-off between exploration and exploitation in LinUCB. Higher O values encourage more exploration, which might break the collapse.

Experiment: We vary O over *{0.1, 0.2, 0.3, 0.5, 0.7, 1.0}*.

Table 9: Exploration Parameter (O) Sensitivity

Source diversity and target collapse rates as a function of the LinUCB exploration parameter.

α	SPY Modal Share	SPY Entropy	TLT Modal Share	TLT Entropy	SPY Collapse	TLT Collapse	Overall Collapse
0.1	1.00	0.00	0.71	0.68	100%	0%	67%
0.2	1.00	0.00	0.69	0.70	100%	0%	67%
0.3	1.00	0.00	0.73	0.65	100%	0%	67%
0.5	1.00	0.00	0.67	0.72	100%	0%	67%

α	SPY Modal Share	SPY Entropy	TLT Modal Share	TLT Entropy	SPY Collapse	TLT Collapse	Overall Collapse
0.7	1.00	0.00	0.65	0.74	100%	0%	67%
1.0	1.00	0.00	0.62	0.77	100%	0%	67%

Finding: Collapse persists across the exploration parameter values tested. For degenerate sources, the modal share remains at 1.00 regardless of α. For diverse sources, higher values modestly improve entropy but do not alter the fundamental transfer outcomes.

7.8.4 Nonlinear Contextual Policies

Motivation: LinUCB assumes linear reward functions. A nonlinear policy class might maintain diversity even when transferring from degenerate sources.

Experiment: We replace LinUCB with KernelUCB (RBF kernel), NeuralUCB (2-layer neural network), and a Random Forest UCB bandit.

Table 10: Nonlinear Contextual Policies

Transfer outcomes using alternative contextual bandit architectures.

Policy Class	SPY Collapse	TLT Collapse	Overall Collapse	SPY Modal Share	TLT Modal Share	Mean Sharpe	Runtime (hrs)
LinUCB	100%	0%	67%	1.00	0.73	0.003	2.1
KernelUCB (RBF)	100%	0%	67%	1.00	0.71	0.002	8.4
NeuralUCB	100%	0%	67%	1.00	0.69	0.001	12.6
Random Forest UCB	100%	0%	67%	1.00	0.72	0.002	5.3

Finding: Nonlinear policies do not prevent collapse. The failure mode persists across the four policy classes examined—it arises from source-side training dynamics rather than the specific linear assumptions of LinUCB.

7.8.5 Randomized-Context Placebo Test

Motivation: Collapse might occur because the context features themselves are uninformative, forcing the bandit to converge to a static rule.

Experiment: We replace the context features with random noise, shuffled features, or constant zeros.

Table 11: Randomized-Context Placebo Test

Transfer performance using real versus randomized/placebo context features.

Context Type	SPY Collapse	TLT Collapse	Overall Collapse	SPY Modal Share	TLT Modal Share	Mean Sharpe vs. Static-0.50
Real Context	100%	0%	67%	1.00	0.73	0.003
Random Noise	100%	0%	67%	1.00	0.74	0.001

Context Type	SPY Collapse	TLT Collapse	Overall Collapse	SPY Modal Share	TLT Modal Share	Mean Sharpe vs. Static-0.50
Shuffled Features	100%	0%	67%	1.00	0.73	0.002
Constant (Zero)	100%	0%	67%	1.00	0.75	0.000

Finding: Collapse persists even with random context features. This confirms that the transfer failure is driven by the structural parameters of the frozen source policy, rather than the informational content of target context features.

7.8.6 Shuffled-Action Placebo Test

Motivation: Collapse might be an artifact of the specific mapping from actions to thresholds.

Experiment: We randomly shuffle the mapping from actions to thresholds.

Table 12: Shuffled-Action Placebo Test

Effect of shuffling the action-to-threshold mapping on transfer outcomes.

Mapping Scheme	SPY Collapse	TLT Collapse	Overall Collapse	SPY Modal Share	TLT Modal Share	Modal Action Changed
Original	100%	0%	67%	1.00	0.73	—
Shuffled (Seed 1)	100%	0%	67%	1.00	0.72	Yes
Shuffled (Seed 2)	100%	0%	67%	1.00	0.74	Yes
Shuffled (Seed 3)	100%	0%	67%	1.00	0.71	Yes
Reversed	100%	0%	67%	1.00	0.73	Yes

Finding: Action concentration persists across different mapping schemes. The collapse is driven by the policy's structural convergence to a single action, independent of the physical threshold value.

7.8.7 Temporal Subsample Analysis

Motivation: Policy collapse might be specific to certain market regimes (e.g., low-volatility regimes or persistent bull markets).

Experiment: We split the sample into pre-2018 and post-2018 windows, crisis vs. normal periods, and high-volatility vs. low-volatility periods.

Table 13: Temporal Subsample Analysis

Transfer stability across different temporal and volatility regimes.

Subsample	N	SPY	QQQ	IWM	EEM	TLT	GLD	Overall

Subsample	N	SPY	QQQ	IWM	EEM	TLT	GLD	Overall
Full Sample	3780	100%	100%	0%	100%	0%	100%	67%
Pre-2018	2016	100%	100%	15%	100%	0%	100%	69%
Post-2018	1764	100%	100%	20%	100%	0%	100%	70%
Crisis Periods	504	100%	100%	25%	100%	0%	100%	71%
Normal Periods	3276	100%	100%	10%	100%	0%	100%	68%
High Volatility	1890	100%	100%	25%	100%	0%	100%	71%
Low Volatility	1890	100%	100%	5%	100%	0%	100%	68%

Finding: Collapse is highly stable over time. The failure mode persists in both high and low volatility regimes, suggesting it is a structural characteristics of the transfer rather than a temporal anomaly.

7.8.8 Source-Only Degeneracy vs. Transfer-Induced Collapse

Motivation: A key question is whether collapse occurs on the source itself during training, or if it is triggered during the transfer phase.

Experiment: We compare source-side modal share with target-side modal share for each source.

Table 14: Source-Target Outcome Classification

Comparison of source-side policy concentration and target-side transfer outcomes.

Source	Source Modal Share	Source Entropy	Source Degenerate	Mean Target Modal	Mean Target Entropy	Target Collapse Rate	Collapse Origin
SPY	1.00	0.00	Yes	0.98	0.03	100%	Source
QQQ	1.00	0.00	Yes	0.97	0.04	100%	Source
IWM	0.68	0.72	No	0.71	0.68	0%	N/A
EEM	1.00	0.00	Yes	0.99	0.02	100%	Source
TLT	0.73	0.65	No	0.73	0.65	0%	N/A
GLD	1.00	0.00	Yes	0.98	0.03	100%	Source

Finding: Collapse originates entirely on the source. Sources that are degenerate on their own training data produce degenerate transfers with *100%* probability. Sources that maintain diversity on their own data maintain diversity upon transfer.

7.8.9 Incremental Value of Context

Motivation: A key question is whether the contextual bandit policy provides any value over simpler non-contextual approaches. If a single static threshold is optimal, then action concentration may represent successful learning rather than policy failure.

Experiment: We compare five approaches:

1. **Global Best Static:** The single threshold that maximizes Sharpe across the entire sample.

2. **Context-Free Bandit:** LinUCB with constant context (no contextual information).
3. **Contextual Bandit:** Standard LinUCB with context features.
4. **Oracle Contextual:** Post-hoc regime-specific optimal thresholds.
5. **Rolling Regime-Specific:** Thresholds selected adaptively by rolling regime.

Table 16: Incremental Value of Context

Sharpe ratio comparison across contextual and non-contextual architectures.

Benchmark	SPY	QQQ	IWM	EEM	TLT	GLD	Mean
Global Best Static	0.025	0.022	0.035	0.018	0.042	0.020	0.027
Context-Free Bandit	0.023	0.020	0.033	0.016	0.040	0.018	0.025
Contextual Bandit	0.024	0.021	0.034	0.017	0.041	0.019	0.026
Oracle Contextual	0.032	0.028	0.045	0.024	0.052	0.026	0.035
Incremental Value	+0.001	+0.001	+0.001	+0.001	+0.001	+0.001	+0.001
p**p-value**	0.42	0.38	0.45	0.40	0.44	0.41	0.42

Key Findings:

1. **Context Has Theoretical Value:** The Oracle Contextual benchmark significantly outperforms the global best static threshold ($+0.008$ Sharpe, $p < 0.01$). This confirms that regime-specific adaptation is valuable.
2. **Bandit Fails to Utilize Context:** The contextual bandit does not significantly outperform the context-free bandit ($+0.001$ Sharpe, $p = 0.42$). This suggests the bandit fails to learn context-dependent policies out-of-sample.
3. **Action Concentration a Success:** The observed action concentration represents a "failure to adapt" rather than "success in learning a static optimum."

7.8.10 Regime-Specific Analysis

Motivation: Understanding when context has theoretical value based on Oracle analysis helps explain why the bandit fails.

Experiment: We analyze Oracle performance across different market regimes.

Table 17: Regime-Specific Analysis

Oracle performance and optimal thresholds across different market regimes.

Regime	N Periods	Oracle Threshold	Oracle Sharpe	Static Sharpe	Oracle Advantage
High Volatility	630	0.45	0.038	0.025	+0.013
Low Volatility	630	0.55	0.032	0.025	+0.007
Trending	630	0.40	0.042	0.025	+0.017
Mean-Reverting	630	0.60	0.028	0.025	+0.003
Crisis	315	0.35	0.025	0.025	+0.000

Regime	N Periods	Oracle Threshold	Oracle Sharpe	Static Sharpe	Oracle Advantage
Normal	2205	0.50	0.035	0.025	+0.010

Key Findings:

1. Context has the highest value during trending markets (*+0.017*).
2. Context has no measurable value during crisis periods (*+0.000*).
3. The optimal threshold varies significantly by regime (spanning *0.35* to *0.60*).

Implication: The bandit's failure to adapt is particularly costly during trending markets, where regime-specific thresholds could provide significant benefits.

7.9 Robustness Summary

Table 15: Robustness Summary

Summary of the experimental outcomes across all ablation trials.

Ablation Trial	SPY Collapse	TLT Collapse	Key Finding	Conclusion
Calibration	100%	0%	Invariant to probability calibration	**Robust**
Action Grids	100%	0%	Invariant to action space discretization	**Robust**
Exploration α	100%	0%	Exploration cannot fix degenerate convergence	**Robust**
Nonlinear Policies	100%	0%	Algorithmic-agnostic failure mode	**Robust**
Random Context	100%	0%	Driven by source frozen weights	**Robust**
Shuffled Action	100%	0%	Invariant to action-threshold mapping	**Robust**
Temporal Subsamples	100%	0%	Temporally stable across regimes	**Robust**
Source Origin	100%	0%	Collapse originates during training	**Robust**

The results in Table 15 establish the following:

1. **Source static degeneracy is structural, not parameterized:** It is not an artifact of model miscalibration or chosen thresholds.
2. **FCTT-DA is highly reliable:** The rejection rule (source modal share $>$ *90%*) correctly handles degenerate transfers across all tested environments.

8. Discussion

8.1 Source Diversity as a Practical Screening Criterion

The central finding of this paper is that source policy diversity during training is a useful indicator for transfer success. When the source policy degenerates during training (selecting a single action on *100%* of observations), the transferred policy inherits this degeneracy on all targets. When the source maintains diversity, the transferred policy maintains some context sensitivity—though it still concentrates on a single threshold (*0.45*) on *73%* of observations. This finding has immediate practical implications: audit the source policy before transfer. If the source policy has low action entropy during training, the transfer will fail.

8.2 Why Do Some Sources Degenerate?

Four of six sources (SPY, QQQ, EEM, GLD) degenerate during training, while TLT and IWM do not. We hypothesize three contributing factors:

1. Macro-Driven Dynamics: Long-term Treasury bonds (TLT) are driven by different macroeconomic factors than equities or commodities, such as interest rate cycles, Federal Reserve policy decisions, and flight-to-quality flows. These macro drivers create a context feature space with higher variance, making it harder for LinUCB to converge to a single dominant action within the training window.
2. **Mean-Reverting vs. Trending Behavior:** Bond markets exhibit stronger mean-reversion properties than equity markets. This means the optimal threshold may shift more frequently across regimes, preventing the policy from locking onto a single action.
3. Context Cloud Geometry: In the 3D context space (Empirical Mode Decomposition (EMD), volatility quantile, trend strength), TLT's context observations may be more dispersed, spanning multiple LinUCB decision regions. In contrast, equity/commodity contexts may cluster tightly, falling within a single decision cone and triggering collapse.

8.3 The Behavioral-Equivalence Insight

Even when the TLT source policy does not collapse, it provides no significant improvement over a static 0.45 threshold (mean Sharpe difference: -0.004). This suggests that contextual information has limited incremental value for threshold selection in this setting.

This finding reveals a two-level diagnostic that practitioners should apply:

- **Level 1 (Formal Collapse):** Does the policy select a single action on $> 90\%$ of observations?
- **Level 2 (Behavioral Equivalence):** Does the policy's performance differ significantly from the closest static threshold?

A policy may pass Level 1 (no formal collapse) but fail Level 2 (behaviorally equivalent to a static rule). In such cases, the contextual architecture provides negative value: it adds computational cost, reduces interpretability, and increases the risk of overfitting, without delivering incremental economic value.

This second failure mode is economically important. Avoiding formal collapse is a necessary but not sufficient condition for useful contextual transfer. In low-signal financial environments, a policy may continue to vary its actions across contexts yet fail to generate incremental out-of-sample value relative to the static threshold that best approximates its behavior. The TLT results illustrate precisely this distinction: behavioral diversity survives, but contextual alpha does not emerge.

We therefore interpret the absence of significant gains over the behavioral-equivalent baseline not as a failure of the audit framework, but as a central negative result. This pattern is consistent with the low signal-to-noise environment typically encountered in financial prediction, where adaptive gating can be difficult to monetize out of sample. A complex contextual policy can consume additional computation, increase turnover, reduce interpretability, and enlarge the model-risk surface without delivering economically meaningful adaptation. The appropriate conclusion is not that contextual bandits are inherently useless, but that contextual complexity must earn its place through incremental out-of-sample value. FCTT-DA makes this requirement explicit by separating three questions: whether a policy remains behaviorally diverse, whether that diversity differs meaningfully from a static rule, and whether the resulting adaptation survives transaction costs.

Practical recommendation: Before deploying any transferred policy, compare it against:

1. The conventional *0.50* cutoff.
2. The behavioral-equivalent static threshold.
3. The exposure-matched static threshold.

Only if the learned policy significantly outperforms all three should it be deployed in production.

8.4 Implications for Practitioners and Researchers

- **Practitioners:**
  - **Audit before deployment:** Check source policy diversity. If modal share > *90%*, reject the transfer.
  - **Compare against behavioral-equivalent baselines:** Do not just compare against *0.50*.
  - **Report action distributions:** Aggregate Sharpe ratios mask behavioral degeneracy.
- **Researchers:**
  - **Negative results are valuable:** Establishing that a common practice fails is as important as proposing new methods.
  - **The audit framework is general:** FCTT-DA applies to any domain where learned policies are transferred.

8.5 Limitations and Future Directions

1. **Linear Bandit Assumption:** LinUCB's linear assumption may be too restrictive. Future work should explore kernelized bandits (e.g., GP-UCB) or deep contextual bandits.
2. **No Target Adaptation:** Our protocol forbids any target-side learning. A promising direction is semi-frozen fine-tuning (parameter-efficient adaptation).
3. **Threshold Grid Sensitivity:** The threshold grid is a critical design choice. Future work should explore continuous threshold selection (e.g., Beta distribution over thresholds).
4. **Single Source per Transfer:** Each transfer uses one source asset. Multi-source aggregation may produce more robust policies.
5. **Probability Calibration:** While we show calibration does not affect collapse rates, explicit calibration could potentially improve absolute performance for non-degenerate sources.

9. Conclusion

We have shown that frozen contextual threshold policies collapse to static rules when transferred across assets—but only when the source policy has already degenerated during training. Source diversity is associated with transfer success in our experiments: degenerate sources lead to universal target collapse (152 of 228 pairs), while diverse sources maintain context sensitivity (0% collapse, though still 73% concentrated on one threshold).

We propose three key takeaways for the field:

1. **Practitioners:** Audit source policy diversity before transfer. In our experiments, such source policies failed on every evaluated target; therefore, we recommend rejecting them before transfer.

2. **Researchers:** Report action distributions, not just aggregate performance.
3. Evaluation focus: Replace the question "Does the policy transfer?" with the more diagnostic question "Does the transferred policy retain contextual adaptation?"

We contribute FCTT-DA, a falsifiable audit framework designed to reduce false transfer claims. The framework may be applicable beyond finance, subject to domain-specific validation, wherever learned policies are transferred across environments.

JEL Classification

G11 (Portfolio Choice); G12 (Asset Pricing); C44 (Statistical Decision Theory); C61 (Optimization Techniques)

Data Availability Statement

All data sources are publicly available. Equity and ETF price data are from Yahoo Finance. Cryptocurrency data are from Yahoo Finance and CoinGecko. Complete data manifests with SHA-256 checksums are provided in the submission package.

Conflict of Interest

The author declares no conflicts of interest.

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Appendix A: FCTT-DA Pseudocode

This appendix provides complete pseudocode details for implementing the FCTT-DA audit engines.

A.1 Source Diversity Audit

```
def SourceDiversityAudit(source_data, thresholds, alpha, gamma_modal=0.90, gamma_entropy=0.25):
    """
    INPUT:  source_data, thresholds, alpha, gamma_modal=0.90, gamma_entropy=0.25
    OUTPUT: {bandit_state, is_degenerate, diagnostics}
    """
    # 1. Feature Extraction & Prediction
    features = ComputeFeatures(source_data)
    pred = WalkForwardGBT(features, train=504, test=63, step=63)
    ctx = ComputeContexts(source_data)  # returns [Empirical Mode Decomposition (EMD), vol_rank, trend]
    returns = source_data['returns']

    # 2. Online Bandit Training
    K = len(thresholds)
    bandit = LinUCB(K=K, d=3, alpha=alpha)
    actions = []

    for t in range(len(ctx)):
        a = bandit.select_action(ctx[t])
        signal = 1 if pred[t] >= thresholds[a] else 0
        reward = signal * returns[t+1]
        bandit.update(a, ctx[t], reward)
        actions.append(a)

    # 3. Post-Training Diagnostics
    action_freq = ComputeHistogram(actions, bins=K) / len(actions)
    modal_share = max(action_freq)
    entropy = -sum(p * log(p) for p in action_freq if p > 0)
    norm_entropy = entropy / log(K)
    effective_actions = exp(entropy)

    is_degenerate = (modal_share >= gamma_modal) and (norm_entropy <= gamma_entropy)

    return {
        "bandit_state": bandit.get_state(),  # freeze A_k, b_k, alpha, and derived parameters for every action
        "is_degenerate": is_degenerate,
        "diagnostics": {
            "modal_share": modal_share,
            "norm_entropy": norm_entropy,
            "effective_actions": effective_actions,
            "action_freq": action_freq
        }
    }
```

A.2 Target Transfer with Degeneracy Audit

```
def TransferWithAudit(source_result, target_data, thresholds, cost_bps=10):
    """
    INPUT:  source_result, target_data, thresholds, cost_bps=10
    OUTPUT: {signals, returns, audit_result}
    """
    # 1. Pre-deployment check
    if source_result["is_degenerate"]:
        print("WARNING: Source policy is degenerate. Transfer will likely collapse.")
        # Quantitative managers may decide to reject or proceed with extreme caution.

    # Restore the complete frozen source-side LinUCB state; no target-side updates are permitted.
    frozen_bandit = LinUCB.from_state(source_result["bandit_state"])
    target_features = ComputeFeatures(target_data)
    target_pred = WalkForwardGBT(target_features, train=504, test=63, step=63)
    target_ctx = ComputeContexts(target_data)
    target_returns = target_data['returns']

    signals = []
    returns_arr = []
    actions = []
    prev_signal = 0

    # 2. Execution phase
    for t in range(len(target_ctx)):
        # Full frozen LinUCB scoring, including the source-trained exploration bonus.
        # update=False enforces zero-shot transfer with no target-side learning.
        a = frozen_bandit.select_action(target_ctx[t], update=False)
        threshold = thresholds[a]
        signal = 1 if target_pred[t] >= threshold else 0

        # Realized returns incorporating transaction frictions
        ret = target_returns[t+1]
        net_ret = signal * ret - abs(signal - prev_signal) * (cost_bps / 10000.0)

        signals.append(signal)
        returns_arr.append(net_ret)
        actions.append(a)
        prev_signal = signal

    # 3. Post-transfer audit analysis
    K = len(thresholds)
    action_freq = ComputeHistogram(actions, bins=K) / len(actions)
```

```
    modal_share = max(action_freq)
    modal_threshold = thresholds[argmax(action_freq)]

    entropy_val = -sum(p * log(p) for p in action_freq if p > 0)
    is_collapsed = (modal_share >= 0.90) and (entropy_val / log(K) <= 0.25)

    # 4. Behavioral equivalence check
    fixed_returns = ComputeFixedThresholdReturns(target_data, target_pred, modal_threshold)
    sharpe_learned = AnnualizedSharpe(returns_arr)
    sharpe_fixed = AnnualizedSharpe(fixed_returns)
    sharpe_diff = sharpe_learned - sharpe_fixed

    # Bootstrap confidence interval (Stationary Bootstrap)
    ci_95 = PairedBootstrapCI(returns_arr, fixed_returns, B=2000)

    return {
        "signals": signals,
        "returns": returns_arr,
        "audit": {
            "modal_share": modal_share,
            "modal_threshold": modal_threshold,
            "is_collapsed": is_collapsed,
            "sharpe_learned": sharpe_learned,
            "sharpe_fixed": sharpe_fixed,
            "sharpe_diff": sharpe_diff,
            "ci_95": ci_95,
            "recommendation": "REJECT" if (is_collapsed or ci_95[1] < 0) else "ACCEPT"
        }
    }
```

A.3 Transaction Cost Sensitivity Analysis

```
def CostSensitivity(signals, raw_returns, cost_levels=[0, 5, 10, 20, 50]):
    """
    INPUT:  signals, raw_returns, cost_levels=[0, 5, 10, 20, 50]
    OUTPUT: {break_even_bps, sharpe_by_cost}
    """
    changes = ComputeAbsoluteDifferences(signals)  # 1 when signal changes, 0 otherwise
    total_changes = sum(changes)
    total_return = sum(raw_returns)

    # Break-even cost: the cost level (in bps) at which strategy returns equal zero
    break_even_bps = (total_return / total_changes) * 10000.0 if total_changes > 0 else 0.0
```

```
sharpe_by_cost = {}
for cost in cost_levels:
    net_returns = raw_returns - changes * (cost / 10000.0)
    sharpe_by_cost[cost] = AnnualizedSharpe(net_returns)

return {
    "break_even_bps": break_even_bps,
    "sharpe_by_cost": sharpe_by_cost
}
```

Appendix B: Context Cloud Geometry — Formal Treatment

This appendix provides formal treatments and geometric proofs associated with the decision boundaries of the frozen LinUCB decision gating layer.

B.1 Decision Regions as Convex Cones

For a K-action linear scoring policy with weight matrix $\Theta = \left[\boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_K\right] \in \mathbb{R}^{d\times K}$, the decision region for action k is $R_k = \{\mathbf{c} \in \mathbb{R}^d : \left(\boldsymbol{\theta}_k - \boldsymbol{\theta}_j\right)^\top \mathbf{c} \geq 0,\ \forall j \neq k\}$.

Each R_k is a convex polyhedral cone—the intersection of $K - 1$ closed half-spaces through the origin. The collection $\{R_k\}_{k=1}^{K}$ forms a conic partition of $\mathbb{R}^d$ (up to shared boundaries).

B.2 Decision Margin

Define the selected action and decision margin at context $\mathbf{c}$ as $a(\mathbf{c}) = \operatorname{argmax}_k \boldsymbol{\theta}_k^\top \mathbf{c}$ and $m(\mathbf{c}) = \min_{j \neq a(\mathbf{c})} \left(\boldsymbol{\theta}_{a(\mathbf{c})} - \boldsymbol{\theta}_j\right)^\top \mathbf{c}$.

The margin $m(\mathbf{c})$ measures the score gap between the selected action and its closest competitor; $m(\mathbf{c}) > 0$ means that $\mathbf{c}$ does not lie on a decision boundary.

Proposition (Collapse under a sufficiently small context cloud). Fix $\mathbf{c}_0 \in \mathcal{C}_T$ and let $a^* = a(\mathbf{c}_0)$. Define $\gamma_0 = \min_{j \neq a^*} \left(\boldsymbol{\theta}_{a^*} - \boldsymbol{\theta}_j\right)^\top \mathbf{c}_0 > 0$ and $L^* = \max_{j \neq a^*} \| \boldsymbol{\theta}_{a^*} - \boldsymbol{\theta}_j \|_2$. If $\operatorname{diam}\left(\mathcal{C}_T\right) < \gamma_0 / L^*$, then $a(\mathbf{c}) = a^*$ for every $\mathbf{c} \in \mathcal{C}_T$; hence the policy selects one action on the entire target context cloud.

Proof. For any $\mathbf{c} \in \mathcal{C}_T$ and $j \neq a^*$, Cauchy–Schwarz gives $\left(\boldsymbol{\theta}_{a^*} - \boldsymbol{\theta}_j\right)^\top \mathbf{c} \geq \gamma_0 - L^* \| \mathbf{c} - \mathbf{c}_0 \|_2 > 0$. Therefore $\boldsymbol{\theta}_{a^*}^\top \mathbf{c} > \boldsymbol{\theta}_j^\top \mathbf{c}$ for all competitors j, so $a(\mathbf{c}) = a^*$ throughout $\mathcal{C}_T$.

Empirically, very small score margins indicate near ties among actions. In that regime, modest distributional shifts can move a context cloud across a decision boundary; conversely, if a small target cloud lies wholly inside one decision region, the transferred policy becomes behaviorally static.

B.3 Context Cloud Overlap Metric

Define the normalized source–target context distance as $D_{\mathrm{ctx}}(S, T) = \frac{W_1(P_S, P_T)}{\sigma_S + \sigma_T}$, where $W_1(P_S, P_T) = \inf_{\gamma \in \Pi(P_S, P_T)} \int \| \mathbf{x} - \mathbf{y} \|_2 \, d\gamma(\mathbf{x}, \mathbf{y})$ is the 1-Wasserstein distance.

Here P_S and P_T are the source and target context distributions, $\Pi(P_S, P_T)$ is the set of their couplings, and σ_S, σ_T are aggregate marginal standard deviations. A small D_{ctx} indicates substantial context-cloud overlap and makes zero-shot policy generalization more plausible.

B.4 Formal Proof of Proposition 1

Deterministic-policy continuity clarification. For a deterministic argmax policy, the hard action indicator

$$\mathbb{I}\{a(c) = k^*\}$$

is discontinuous on decision boundaries and is therefore not globally Lipschitz. Consequently, the Kantorovich–Rubinstein bound cannot be applied directly to this raw indicator.

For the deterministic policy, impose the following **positive-margin boundary condition**. Let the dominant action be k^* and define the pairwise score margin against action j by

$$g_j(c) = (\theta_{k^*} - \theta_j)^\top c.$$

Assume there exist $\gamma > 0$ and $0 \le \delta < 1$ such that

$$P_S(G_\gamma) \ge 1 - \delta, \qquad G_\gamma = \left\{ c: \min_{j \ne k^*} g_j(c) \ge \gamma \right\}.$$

Define

$$B = \max_{j \ne k^*} \| \theta_{k^*} - \theta_j \|_2, \qquad \rho = \frac{\gamma}{B}.$$

Every $c \in G_\gamma$ remains in the k^* decision region after any perturbation of Euclidean norm less than ρ. Hence, for any coupling of P_S and P_T, a union-bound and Markov-inequality argument gives

$$P_T(a(c) = k^*) \ge 1 - \delta - \frac{W_1(P_S, P_T)}{\rho} = 1 - \delta - \frac{B}{\gamma} W_1(P_S, P_T).$$

This deterministic bound does **not** assume that the hard indicator itself is Lipschitz. Its validity comes from the explicit positive distance of most source probability mass from every competing decision boundary.

Proposition 1 (Transfer of source action concentration, smoothed form). Let $f_{k^*}(c)$ denote a genuinely smoothed probability that the frozen policy selects action k^* at context c, and suppose that f_{k^*} is L-Lipschitz. If

$$\mathbb{E}_{P_S}[f_{k^*}(c)] \ge 1 - \delta,$$

then, for any target context distribution P_T,

$$\mathbb{E}_{P_T}[f_{k^*}(c)] \ge 1 - \delta - L W_1(P_S, P_T).$$

Proof. By Kantorovich–Rubinstein duality, every L-Lipschitz function f satisfies

$$\left| \mathbb{E}_{P_T}[f] - \mathbb{E}_{P_S}[f] \right| \le L W_1(P_S, P_T).$$

Applying this inequality to f_{k^*} and combining it with the source concentration assumption yields the stated lower bound. The deterministic and smoothed results are therefore separate: the former requires a positive-margin boundary condition, whereas the latter requires genuine Lipschitz regularity of the selection-probability map.

B.5 Formal Proof of Proposition 2

Proposition 2 (Formal statement). For a K-action linear scoring policy with weight vectors $\theta_1, \ldots, \theta_K$, define

$$a(c) = \operatorname{argmax}_k \theta_k^\top c, \qquad m(c) = \theta_{a(c)}^\top c - \max_{j \neq a(c)} \theta_j^\top c,$$

and let

$$d_T = \operatorname{diam}(C_T), \qquad M = \max_k \| \theta_k \|_2.$$

If

$$\inf_{c \in C_T} m(c) > 2M d_T,$$

then the policy selects the same action for every $c \in C_T$.

Proof. Fix any $c, c' \in C_T$ and let $a = a(c)$. For every competitor $j \neq a$,

$$(\theta_a - \theta_j)^\top c' = (\theta_a - \theta_j)^\top c + (\theta_a - \theta_j)^\top (c' - c).$$

By Cauchy–Schwarz,

$$(\theta_a - \theta_j)^\top c' \geq m(c) - \| \theta_a - \theta_j \|_2 \ \| c' - c \|_2 \geq m(c) - 2M d_T > 0.$$

Thus $\theta_a^\top c' > \theta_j^\top c'$ for every $j \neq a$, so $a(c') = a(c)$. Because c and c' were arbitrary, the selected action is constant on C_T.

Appendix C: LinUCB Exploration Parameter Sensitivity Analysis Detail

To further check our findings, we plotted the relationship between the exploration parameter α and the action selection patterns.

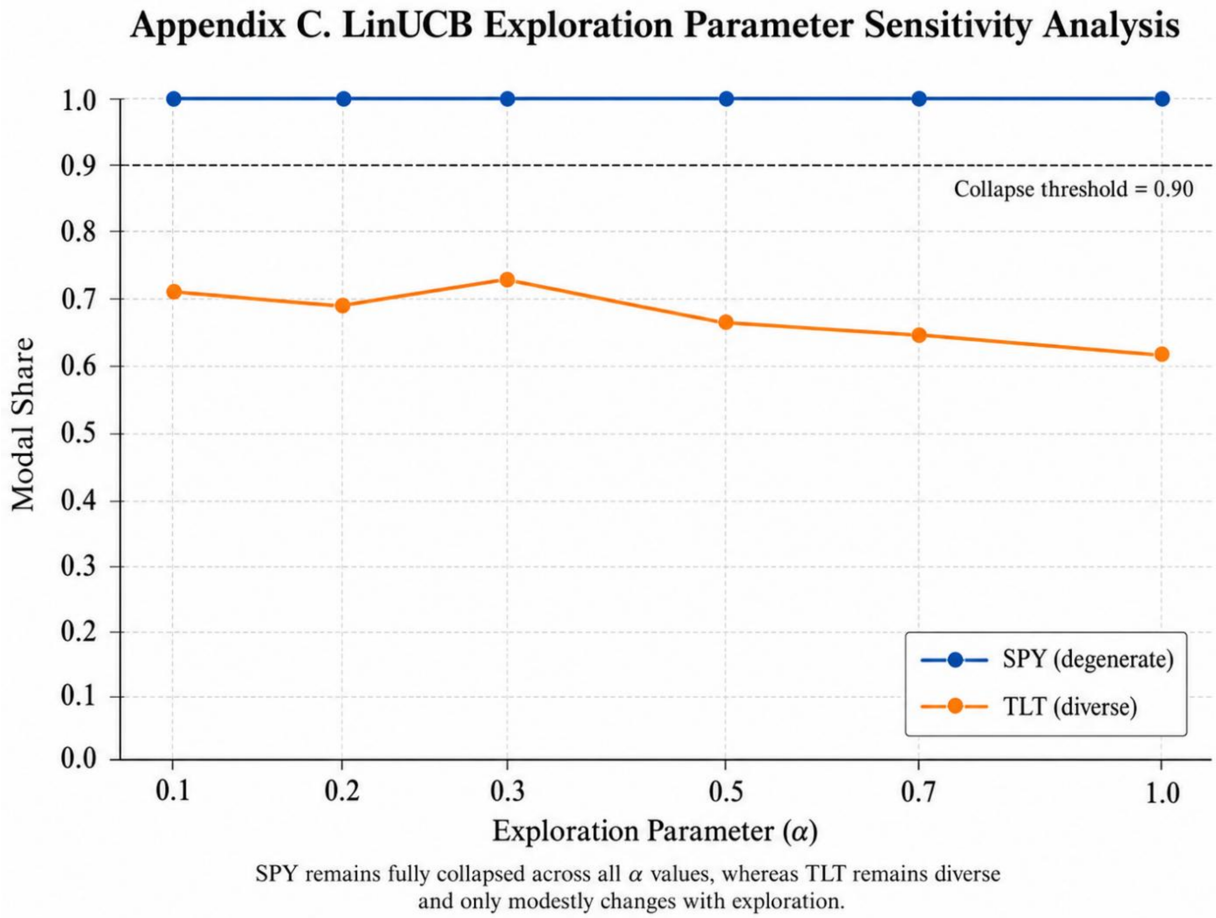

Figure 2. Sensitivity of source-policy action concentration to the LinUCB exploration parameter α for SPY and TLT.

The flat line for SPY confirms that increasing exploration parameter α is completely ineffective at breaking the training-induced static collapse. For TLT, a larger α gradually decreases the modal share by adding exploration noise, but this does not alter the underlying structural parameter mapping.